De Montfort University
Beng (Hons) Mechanical Engineering – Year 3
Module: Mechanical Energy Analysis - ENGE3002
Experiment: Diesel Engine

Coursework: The Performance & Emissions of a Diesel Engine

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Introduction:

Compression ignition (diesel) engines are widely used in automotive and industrial applications due to their high thermal efficiency and durability. However, increasingly strict emissions regulations have made it necessary to better understand how engine operating conditions influence efficiency, fuel consumption, and exhaust emissions. Experimental engine testing provides a practical means of analysing these effects and evaluating engine behaviour under controlled conditions.

In this coursework, a laboratory-based experimental investigation was carried out on a four-cylinder compression ignition engine operating at two fixed engine speeds of 2300 rpm and 2800 rpm. At each speed, the engine load was varied across multiple steady-state test points. Key operating parameters were measured, including fuel and air flow rates, cooling water temperatures, exhaust temperatures, and gaseous emissions, allowing the impact of load and speed on engine performance to be examined.

The primary objective of the experiment is to investigate the relationship between engine load, represented by brake mean effective pressure (BMEP), and engine performance and emissions. Engine efficiency is evaluated using brake thermal efficiency and brake specific fuel consumption (BSFC), while combustion and mixture behaviour are assessed through air-fuel ratio. Exhaust emissions analysis includes measurements of carbon monoxide (CO), carbon dioxide (CO_2), nitrogen oxides (NO_x and NO), total hydrocarbons (THC), and oxygen (O_2).

Additionally, an energy balance is performed to assess how much fuel energy is converted into useful brake power, heat rejected to the cooling system, energy carried away in the exhaust gases, and other losses. Through this analysis, the coursework aims to provide insight into the performance and emissions characteristics of a modern diesel engine operating under steady-state conditions.

Euro Emissions Standards

Introduction:

Euro emission standards are applied by European Union member countries and others, including the UK, to regulate exhaust emissions such as carbon monoxide (CO) and nitrogen oxides (NO_x) to improve air quality and reduce pollution, with stricter limits being implemented over time. The time between standards is generally between four and six years, allowing for steady, but significant, reduction in vehicle emissions over time. These regulations are applied on all new road vehicles; cars, vans, trucks, buses and motorcycles, and must comply with the latest Euro standard which is currently Euro 6 but will be Euro 7 effective November 29, 2026, for new vehicles. Heavy-duty vehicles like lorries and buses have their own standards and Non-Road Mobile Machinery (NRMM) engines in equipment like generators are also regulated.

There are many potential strategies to reducing emissions from engines. Different engines have different advantages and fit better for different scenarios. The reduction in emissions is highly influenced by many factors such as electricity & heat production (~35-40%), industry (steel, cement, chemicals) (~20-25%), transport (~20-25%). Most of the transport is road transport and most emissions caused by road transport is by passenger cars (~40-45%), followed by heavy-duty trucks and buses (~30-35%), then vans (~15-20%).

Euro emission regulations are important especially in London due to emission fees for vehicles that do not meet the criteria in Ultra-Low Emission Zones (ULEZ). ULEZ requires Euro 6 standards for Diesel and Euro 4 for petrol. Vehicles manufactured after June 2006 for petrol and after September 2015 for diesel are generally ULEZ-compliant. This has led to manufacturers designing new technologies for diesel engines to meet the stricter limits on NO_x emissions, these technologies require more complex manufacturing processes and can increase production cost.

Furthermore, the European Union has implemented various incentives to encourage companies to exceed emission standards, this is important since popular brands such as Volkswagen and BMW among others are based in Germany. Most European countries provide tax breaks, purchase rebates, free charging, and reduced road toles for zero emission vehicles, Automakers can pool carbon emissions credits with leaders like tesla to avoid penalties that could total hundreds of millions of euros. Carbon emissions credits are tradable allowances earned when exceeding manufacturer regulatory targets, which automakers like tesla sell to other car companies who can then meet their targets although technically exceeding emissions.

Euro 1-6:

Euro 1 (1992/1993): Euro 1 standards marked a significant step towards improving air quality by limiting the amount of carbon dioxide (CO_2) emitted by cars. The main changes implemented to meet these standards include catalytic converters (fig. 1.) required for petrol cars to reduce CO emissions to (2.72 g/km petrol/diesel), unleaded petrol becoming mandatory for all petrol engines, hydrocarbons (HC) + NO_x limits of (0.97 g/km) set for petrol/diesel and finally a particulate matter limit of (0.14 g/km) for diesel only.

Leaded petrol contained tetraethyl lead (TEL), a toxic compound added to the fuel to boost octane, prevent engine knocking (pre-ignition), and protect valve seats in older engines, allowing for more powerful engines. This additive, however, released harmful lead particles into the environment, causing sever public health issues.

Table 1 Euro 1 Emissions Standard (car)

Pollutant	Petrol Limit	Diesel Limit
Carbon Monoxide (CO)	2.72 g/km	2.72 g/km
Hydrocarbons + Nitrogen Oxides (HC + NO_x)	0.97 g/km	0.97 g/km
Particulate Matter (PM)	No limit	0.14 g/km

Both petrol and diesel had the same limits at this except for PM which was exclusively regulatory for diesel cars.

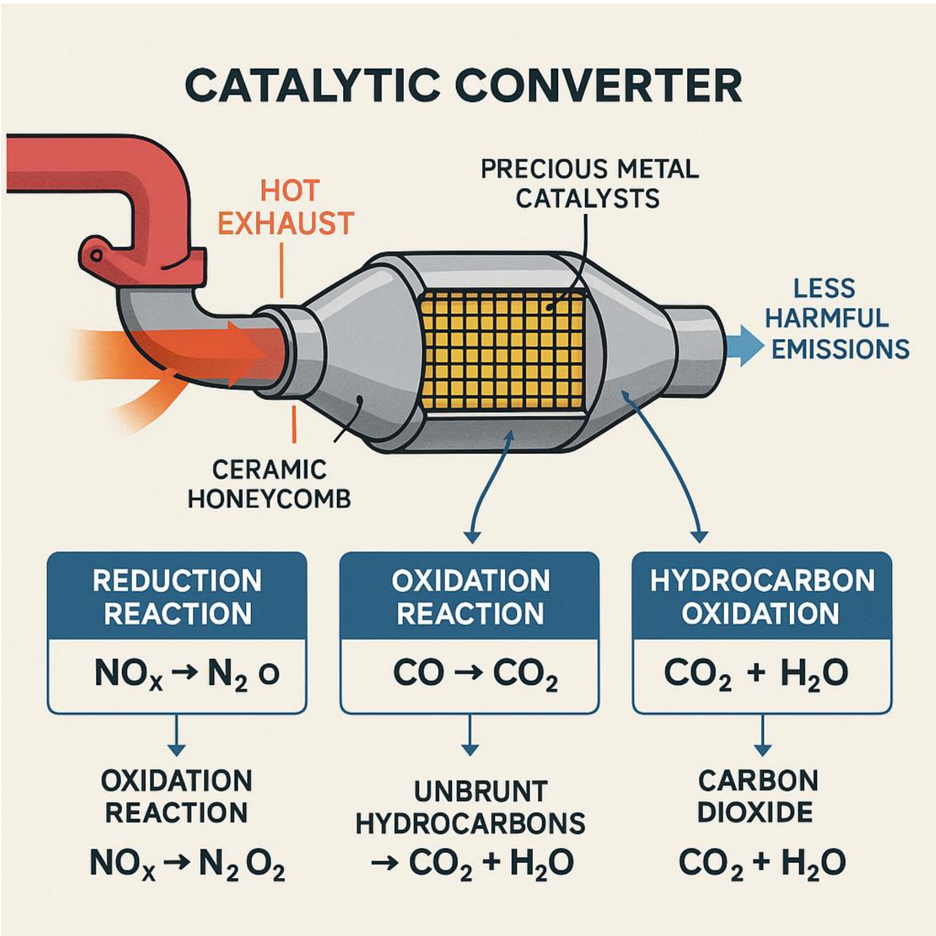

Figure 1 Catalytic Converter

Euro 2 (1996/1997): This regulation was applied to all new registered cars in the EU from January 1, 1997. It was the first standard to introduce separate emissions limits for petrol and diesel vehicles.

Table 2 Euro 2 Emissions Standard (car)

Pollutant	Petrol Limit	Diesel Limit
Carbon Monoxide (CO)	2.20 g/km	1.0 g/km
Hydrocarbons + Nitrogen Oxides (HC + NO_x)	0.50 g/km	0.70 g/km
Particulate Matter (PM)	No limit	0.08 g/km

Due to these stricter standards, including the significant difference from Euro 1 PM limit (0.14 g/km) to Euro 2 (0.8 g/km), manufacturers utilised more sophisticated engine management systems to precisely control the combustion process and minimise the creation of pollutants. Manufacturers also improved the already mandatory catalytic converters, improving their efficiency and durability to meet the stricter standards. Advancements in fuel injection systems, such as widespread use of direct injection, helped optimise the air-fuel mixture for more complete combustion, reducing emissions. The strict PM limit for diesel engines demanded automakers to refine diesel engine design to reduce the amount of soot produced during combustion.

Euro 3 (2000/2001): Effective for all new car registrations from January 1, 2001. A key change was the introduction of separate, specific limits for NO_x for both petrol and diesel engines and the removal of the engine warm-up period during testing. Engine warm-up allows for the catalytic converter to reach its optimal operating temperature, removing this grace period exposed the higher initial emissions caused by cold starts, which meant stricter limits and a push for manufacturers to reduce emissions during cold starts.

Table 3 Euro 3 Emissions Standards (car)

Pollutant	Petrol Limit	Diesel Limit
Carbon Monoxide (CO)	2.3 g/km	0.64 g/km
Total Hydrocarbons (THC)	0.2 g/km	0.56 g/km (combined HC + NO_x)
Nitrogen Oxides (NO_x)	0.15 g/km	0.50 g/km
Particulate Matter (PM)	No limit	0.05 g/km

The CO emissions increased to (2.3 g/km) from the previous Euro 2 standard of (2.2 g/km), this is not to increase emissions but was because of the new testing implemented where engine warm-up was removed from testing. The higher number was to take into consideration this factor. For the first time, diesel had to meet a specific NO_x target (0.50 g/km) which was implemented to prevent manufacturers from trading lower HC emission values for higher NO_x emissions.

Euro 4 (2005/2006): Application began January 1, 2006. Euro 4 was the first standard where Diesel Particulate Filters (DPFs) (*Fig. 2.*) began appearing in production cars. These ceramic honeycomb structures trap fine soot particles as part of the exhaust system, which are then burned off through a "regeneration" process, generally requiring driving above a certain speed and time. Exhaust Gas Recirculation (EGR) was improved in production cars; it is an emissions control technology used in both petrol and diesel engines to reduce NO_x

emissions by redirecting a controlled portion of an engine's exhaust gases back into the combustion chamber.

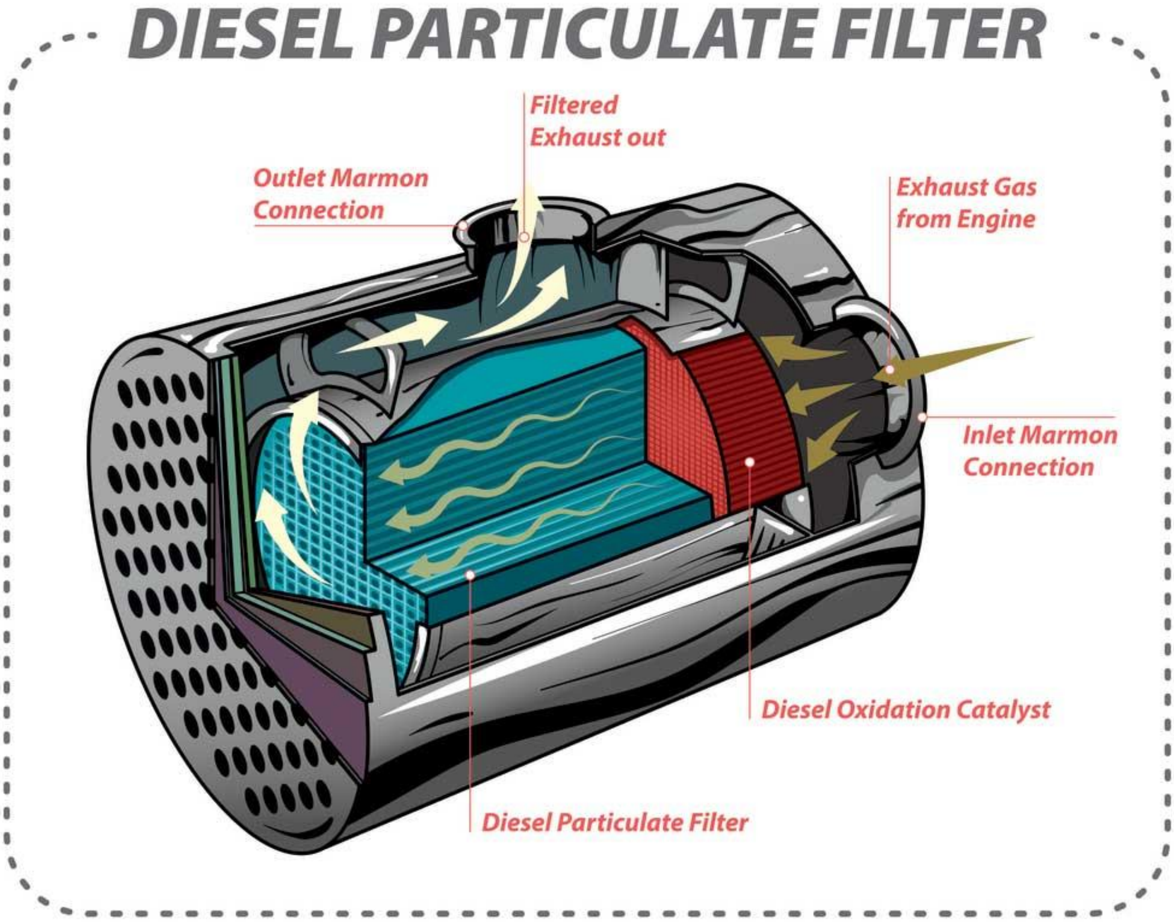

Figure 2 Diesel Particulate Filter

Furthermore, a separate limit on Non-Methane-Hydrocarbons (NMHC) was put in place for petrol cars for the first time in Euro emissions standards. The limit of (0.068 g/km) was introduced as HCs and NMHCs pose different risks environmentally; methane (major HC) is a potent greenhouse gas but relatively unreactive in the lower atmosphere and does not contribute significantly to local air quality issues like smog. NMHC includes gases like benzene, ethane, and propane which are highly reactive and are the primary precursors for photochemical smog and can react with NO_x in sunlight producing ground-level ozone.

Ground-level ozone contributes to global warming, lowers the production of staple crops and can inhibit photosynthesis among other environmental effects. It can also cause respiratory issues for humans, worsen conditions such as asthma and is linked to premature deaths. It also causes material degradation such as causing rubber to crack.

Table 4 Euro 4 Emissions Standard (car)

Pollutant	Petrol Limit	Diesel Limit
Carbon Monoxide (CO)	1.0 g/km	0.50 g/km
Total Hydrocarbons (THC)	0.1 g/km	0.30 g/km (combined HC + NO_x)

Nitrogen Oxides (NO_x)	0.08 g/km	0.25 g/km
Particulate Matter (PM)	No limit	0.025 g/km
Non-Methane HC (NMHC)	0.068 g/km	No limit

Euro 4 introduced a 50% reduction in nearly all pollutant limits, like THC limit being further restricted to (0.1 g/km) for petrol cars. To achieve these specific targets, several key technologies were standardised:

- **Three-way Catalytic Converters (petrol):** Refined catalytic converters to oxidise unburned hydrocarbons into water and carbon dioxide more efficiently, especially during the "cold start" period.
- **Diesel Oxidation Catalysts (DOC):** Used in diesel engines to break down organic fraction of particulates and hydrocarbon gases.
- **On-Board Diagnostics (OBD):** OBD systems that monitor the efficiency of the catalyst in reducing HC and NO_x.

Euro 5 (2009/2011): Mandatory by January 1, 2011. Focus shifted heavily onto PM, especially for diesels.

Table 5 Euro 5 Emissions Standard (cars)

Pollutant	Petrol Limit	Diesel Limit
Carbon Monoxide (CO)	1.0 g/km	0.50 g/km
Total Hydrocarbons (THC)	0.1 g/km	0.23 g/km (combined HC + NO_x)
Nitrogen Oxides (NO_x)	0.06 g/km	0.25 g/km
Particulate Matter (PM)	0.005 g/km	0.005 g/km
Particulate Number (PN)	No limit	$6.0 \times 10^{11}/km$
Non-Methane HC (NMHC)	0.068 g/km	No limit

A new Particulate Number (PN) limit (6.0×10^{11} /km) was introduced for diesel cars, petrol cars then had to follow PM regulations like diesel cars in the past for the first time. Manufacturers had to further develop and standardise more pollutant treatment and engine management technologies:

- **Diesel Particulate Filters (DPF):** DPFs became effectively mandatory for all new diesel cars. Filters trap 99% of soot particles reducing PM emissions by roughly 80% from Euro 4 standards (0.025 g/km) to Euro 5 standards (0.005 g/km)
- **PN limits:** This Euro standard was the first time regulators measured not just the mass of the soot, but the actual number of fine particles emitted per kilometre.
- **Gasoline Direct Injection (GDI) controls:** Euro 5 was the first standard to apply PM limits to petrol cars, specifically those using direct injection technology.
- **Advanced Variable Geometry Turbochargers (VGT):** Used to improve engine efficiency and reduce engine-out emissions before they even reached the exhaust filters.
- **Standardised OBD II:** Systems were further enhanced to monitor the efficiency of the DPF and alert drivers if the filter became clogged and required regeneration.

The PN limit was introduced in the Euro 5b standard effective (2013) primarily because mass-based measurements (PM) had reached their technical limits and failed to capture the most dangerous types of pollution. The measurement accuracy of PM is limited, since PM limits became so low, even minor background noise or weighing errors could lead to inaccurate results. Counting individual particles (PN) proved to be significantly more sensitive than weighing them (PM), making it a much more reliable way to certify that a vehicle is truly clean.

While larger particles make up most of the weight of exhaust particles, most of the quantity of particles are ultrafine (smaller than 0.1 micrometres). These ultrafine particles can bypass the lungs natural filters, enter the blood stream and travel to vital organs. Since, they have a much larger surface area relative to their size they can carry and transport more toxic chemicals into the body and are linked to risks of heart disease, Alzheimer's, respiratory mortality among many. Introducing a PN limit forced manufacturers to ensure the most effective filtration technology available; the strict limit of 600 billion particles per kilometre could only be met using DPF which traps 99% of soot.

As petrol cars moved to Direct Injection for better fuel efficiency, they began producing high numbers of PM like diesel engines, making a PM limit necessary for them as well.

Euro 6 (2014/2015): Effective from September 1, 2015, for all new registrations. Petrol limits remained largely consistent with Euro 5 while diesel limits were brought closer to petrol levels. Euro 6 focused on a dramatic reduction in NO_x emissions, particularly from diesel engines and the introduction of PN limits for petrol cars. Although, Port Fuel Injection (PFI) used in traditional petrol cars produce particle matter numbers and mass well below the limit, which is why Euro 5 standards and above, limit specifically direct injection (GDI) petrol engines.

Table 6 Euro 6 Emissions Standard (cars)

Pollutant	Petrol Limit	Diesel Limit
Carbon Monoxide (CO)	1.0 g/km	0.50 g/km
Total Hydrocarbons (THC)	0.1 g/km	0.17 g/km (combined HC + NO_x)
Nitrogen Oxides (NO_x)	0.06 g/km	0.08 g/km
Particulate Matter (PM)	0.005 g/km (GDI/DI)	0.005 g/km
Particulate Number (PN)	$6.0 \times 10^{11}/km$ (GDI/DI)	$6.0 \times 10^{11}/km$
Non-Methane HC (NMHC)	0.068 g/km	No limit

Core technologies for Euro 6:

- **Selective Catalytic Reduction (SCR):** This system injected AdBlue into the exhaust stream. A chemical reaction in the catalyst converts NO_x into harmless water and nitrogen
- **Lean NO_x Traps (LNT):** Used in smaller diesel engines, adsorbers store NO_x during normal driving and periodically burn it off into nitrogen during brief fuel-rich cycles.
- **Gasoline Particulate Filters (GPF):** To meet PN count limit for petrol cars, many manufacturers began fitting filters to petrol exhausts, like DPFs used in diesels.

Evolution of Euro 6 (2017-2026):

Euro 6 has several sub-stages that changed how vehicles were tested rather than just the limits themselves

- **Euro 6c (2017):** Laboratory test changed from NEDC to more realistic WLTP (Worldwide Harmonised Light Vehicle Test Procedure).
- **Euro 6d-TEMP (2019):** Introduced Real Driving Emissions (RDE) tests, where cars were tested on actual roads to match reality.
- **Euro 6d (2021):** Real world NO_x emissions were required to be much closer to the laboratory limit.
- **Euro 6e-bis (April 2026):** in 2026, the UK plans to mandate this standard, which specifically targets more accurate CO2 and pollutant measurements for Plug-in Hybrids (PHEVs).

In 2026, Euro 6 remains the gold standard for urban access. Diesel vehicles must be at least Euro 6 compliant to enter London's ULEZ zones and various UK Clean Air Zones (CAZ) without a relatively high daily charge (£12.50 for ULEZ). Diesel cars that meet the latest standards (Euro 6d) avoid specific tax surcharges in the UK.

Emerging Technologies

This section reviews emerging compression ignition engine technologies and fuels aimed at reducing and regulating greenhouse gas emissions. A broad overview of current approaches is presented, followed by a detailed discussion of three particularly influential developments.

Table 7 Primary efficiency/emissions values of a conventional modern diesel engine, operating under steady conditions.

	Diesel
NO_x	~4-8 g/kWh
PM (Particulate Matter)	~0.05-0.1 g/kWh
Brake Thermal Efficiency	~40-45%
BSFC (Brake Specific Fuel Consumption)	~190-220 g/kWh

Compression-Ignition Engine Technologies:

- **HCCI (Homogeneous Charge Compression Ignition):** Fuel and air are fully premixed before compression; ignition spontaneously occur due to temperature and pressure. Without direct timing control. Not suitable for full-range road vehicles.
- **PCCI (Premixed Charge Compression Ignition):** Partial premixing strategy between diesel spray combustion and HCCI. Limited high-load operation. Light/medium-duty diesel cars.
- **RCCI (Reactivity Controlled Compression Ignition):** Dual fuel combustion using low-reactivity fuel (petrol/ethanol) and high-reactivity fuel (diesel) to control ignition. Heavy-duty trucks, stationary power generation and large marine engines.

- **Advanced High-Pressure Common Rail Injection:** Fuel injection at 2500-3000 bar with multiple injection events. High costs/mechanical stress. Passenger cars, heavy-duty trucks, rail/marine auxiliary engines.
- **Advanced Air Management (Turbocharging + EGR):** Improved control of intake air density and exhaust gas recirculation. Fits in all modern diesel engines, mandatory for emissions compliance (universally applicable system).
- **Exhaust Aftertreatment Systems (DOC + DPF + SCR):** Chemical conversion of pollutants post-combustion for controlled emissions. Universally applicable, used in most modern engines such as road vehicles, rail, marine engines and stationary power (used to meet regulations).

Alternative Fuels:

- **Biodiesel (FAME):** Fits well in urban fleets, taxis, municipal fleets and stationary generators but not ideal for cold climates or marine use.
- **HVO (Hydrotreated Vegetable Oil):** Heavy-duty trucks, rail, marine engines, drop-in replacement. Potentially demand in immediate deployment sectors.
- **Synthetic E-Diesel (Power-to-Liquid):** Emissions of pollutants near zero except NO_x which is similar diesel. Marine shipping, rail, aviation-adjacent sectors but not mass-market automotive.
- **Dual-Fuel Diesel-Gas (LNG, CNG):** Cargo ships, long-haul trucks, stationary engines.
- **Hydrogen Compression Ignition:** Near zero emissions of pollutants except NO_x which is like diesel. Research engines, prototype heavy-duty systems but not passenger cars or marine yet.

Highest Potential:

Table 8 Approximate steady state values for RCCI and comparison against traditional modern diesel

	RCCI (typical range)	RCCI vs Diesel
NO_x	~0.1-0.5 g/kWh	~90-98% ↓
PM (Particulate Matter)	< 0.01 g/kWh	~90-99% ↓
Brake Thermal Efficiency	~45-50%	~5-10 pts ↑
BSFC (Brake Specific Fuel Consumption)	~170-190 g/kWh	~10-15% ↓

RCCI: Universities and researchers have found NO_x reduction of ~90-98% (*Tab. 9.*), PM near zero (often below detection), Brake thermal efficiency increases to ~45-50% (*Tab. 9.*). Combustion phasing is controlled chemically not mechanically.

Conventional diesel engines control combustion phases mechanically. This requires, injection timing, injection pressure, injection rate shape and EGR (exhaust-gas-recirculation). Ignition is still governed by spray mixing and local conditions; fuel auto-ignites in localised rich zones, large temperature gradients, high peak temperatures and uncontrolled ignition delay variability. This causes NO_x, soot and variations per cycle.

Advantages:
Chemical control (RCCI) controls ignition by fuel reactivity, mixture homogeneity and chemical kinetics. Two fuels are used: low-reactivity (petrol/ethanol, pre-mixed) and high-reactivity (diesel, ignition trigger). This gives predictable combustion phasing and reduced sensitivity to variations in combustion cycles. Smooth heat release also helps reduce NO_x emissions as NO_x forms exponentially above ~1800 K. Diesel flames exceed 2000 K but in RCCI peak temperatures are often <1800 K. Similarly soot formation is nearly eliminated in RCCI due to its lean mixture and no rich pockets during combustion, where traditional diesels are fuel-rich and oxygen poor operating with high-temperature zones.

Problems:
Requires two fuels. This means two tanks, two fuel lines, two injectors or port + direct injection. This is a problem for passenger cars, the main transport contributor of emissions, due to its impracticality. Fleet refuelling is manageable and for heavy-duty it is possible but expensive and complex to manufacture and buy which is important for businesses that invest in heavy-duty vehicles/machinery. Furthermore, for RCCI, the control of cylinder wall temperature, air temperature, fuel ratio, injection timing, intake pressure. Small errors cause misfire, knock, late combustion and more issues. More parts and complexity mean worse long-term reliability.

Conclusion:
Overall, RCCI achieves ultra-low emissions by chemically controlling combustion phasing through fuel reactivity rather than mechanically though injection timing, resulting in low peak temperatures that suppress both NO_x and soot formation at the source; however, its reliance on dual-fuel infrastructure, high control complexity, poor cold-start behaviour and instability during changing-variable operation currently limit widespread commercial adoption.

Table 9 Approximate steady state values for PCCI and comparison against traditional modern diesel

	RCCI (typical range)	RCCI vs Diesel
NO_x	~1-3 g/kWh	~40-70% ↓
PM (Particulate Matter)	~0.01-0.03 g/kWh	~50-80% ↓
Brake Thermal Efficiency	~41-45%	≈ same
BSFC (Brake Specific Fuel Consumption)	~190-210 g/kWh	~0-10% ↓

PCCI: This is the one manufacturer want for passenger cars. PM ~50-80% (*Tab. 9.*), NO_x ~40-70% (*Tab. 9.*) and maintains diesel efficiency. PCCI is partially premixing diesel with air before ignition. This is done by early injection, allowing time for fuel-air mixing and still relying on traditional compression ignition.

In traditional diesel engines, diffusion combustion causes fuel-rich pockets which can cause incomplete combustion due to poor mixing. This leads to the production of CO, HC (hydrocarbons), methane, and PM. As previously discussed in the RCCI section, NO_x depends on peak flame temperatures and more so, residence time at high temperatures. Traditional diesel engines have high temperatures further leading to emissions productions.

Advantages:
Since PCCI reduces rich zones and spray core diffusion flames, diffusion combustion is reduced. PCCI lowers peak temperatures and releases heat more homogeneously than traditional diesels. This means a reduction in the production of emissions gases and particulate matter.

Although PCCI introduces a new strategy, the core mechanism of a traditional diesel stays; combustion is still timed, ignition is still controlled by injection. This means combustion phasing can be controlled to the same extent as traditional diesels to cause ignition to happen exactly when the piston is at the top, this pushes the piston down through its entire power stroke, capturing the maximum amount of energy. The premixed fuel will burn faster than a standard diesel flame, which means fast concentrated combustion finishes quickly. This prevents a flame lingering late in the stroke, which would waste energy by heating up the cylinder walls instead of producing work onto the piston.

Phasing combustion near TDC keeps traditional torque. The engine's computer can adjust injection timing for every single stroke, the engine responds smoothly to the driver's input, keeping traditional driveability. By reducing heat loss and timing the burn at TDC, less fuel is required to generate the same amount of power as a traditional diesel, this improves mpg.

Disadvantages:
At high load using pure PCCI, the early injection causes excessive pressure rise, causing knock-like combustion, noise and mechanical stress. So PCCI is typically partial-load only and blended with conventional diesel at high load.

Conclusion:
PCCI acts as a bridge technology, subtly improving diesel engines without demanding a different ecosystem to use it. Thermal efficiency is improved slightly, with reductions of emissions by up to ~80%. Manufacturers can use this technology to meet emissions standards without huge expense or losing customers due to impracticality. This technology holds realistic potential to be adopted widely.

Table 10 Approximate steady state values for Advanced Diesel Systems + Renewable Drop-in Fuels and comparison against traditional modern diesel

	RCCI (typical range)	RCCI vs Diesel
NO_x	~3-7 g/kWh	~0-20% ↓
PM (Particulate Matter)	~0.02-0.06 g/kWh	~20-50% ↓
Brake Thermal Efficiency	~40-44%	≈ same
BSFC (Brake Specific Fuel Consumption)	~195-220 g/kWh	≈ same

Advanced Diesel Systems + Renewable Drop-in Fuels: This combination of technologies and fuels covers:

Hardware:

- 2500-3000 bar common rail.
- Multi-hole injectors.
- Variable geometry turbochargers.
- High EGR capability.

Aftertreatment:
- DOC.
- DPF.
- SCR (AdBlue).

Control:
- Cylinder pressure-based combustion control.
- Adaptive Injection strategies.

Modern heavy-duty diesels can achieve NO_x reduction of >90% and PM reduction of >95%. Not through clean combustion alone but system-level control. The engine produces dirty exhaust; the system cleans it before release. This is why advanced combustion is less urgent in this sector.

Heavy-duty diesels are used in road transport (trucks, construction, buses…), off highway & industrial (tractors, excavators, forklifts…) and stationary engines (backup generators, remote power generation…). These engines run at steady load, don't tolerate experimental combustion due to several factors like money, time and convenience. Drop-in fuels, are fuels that are compatible with diesel fuel standards, have similar properties and can be use in an engine with no hardware changes. Heavy-duty engines reject clever fuels that require complex systems and new infrastructure because reliability beats theoretical efficiency.

Advantages:
Drop-in fuels keep uptime unaffected, don't risk reliability or maintenance intervals and are widely available. Manufacturers need minimal risk with improved results. HVO and synthetic diesel have higher cetane than fossil diesel, this means shorter ignition delay, more controlled combustion etc. Heavy-duty engines are designed around cetane behaviour, which allows drop-in fuels to thrive. Drop-in renewable diesels have no aromatics, no sulphur and fewer impurities; lower soot, easier DPF regeneration, longer oil life.

Cetane number of HVO ~70-90. Near zero sulphur. CO_2 produced by HVO: ~60-90% reduction, synthetic diesel (renewable electricity): up to ~90%.

For heavy-duty use, every hour matters, fuel use dominates emissions not warm up phases and lifecycle CO_2 savings are huge.

Disadvantages:
Passenger cars experience lots of cold starts, short trips, transient loads where emissions are dominated by the warmup phase. Drop-in fuels still help but the benefits are diluted, the cost is harder to justify, and hybridisation often beats fuel switching.

Conclusion:

Renewable drop-in fuels are a low risk, emissions reducing strategy that has real world potential to be used widely by heavy duty diesel engines. The reduction in emissions is significant, with a possible reduction of 50% (*Tab. 10.*) in PM

		Engine Speed 1 Results					Engine Speed 2 Results				
Target RPM		2300	2300	2300	2300	2300	2800	2800	2800	2800	2800
Measured Performance Values	**Units**	Test A	Test B	Test-C	Test-D	Test-E	Test A	Test B	Test-C	Test-D	Test-E
Engine Brake Power	kW	44.3	29.2	12.2	7.2	4.3	53.2	34.5	15.8	8.1	4.1
Engine Torque	Nm	180.0	125.0	49.0	29.0	17.0	178.0	178.0	178.0	178.0	178.0
Engine Speed	rpm	2343	2227	2379	2382	2415	2853	2853	2853	2853	2854
Engine Cooling Water Flow Rate	L/min	32.0	32.0	32.0	32.0	32.0	39.0	39.0	39.0	39.0	39.0
Relative Humidity	%	37.0	37.0	38.0	38.0	38.0	40.0	38.0	36.0	37.0	37.0
Engine Water Temp In	°C	60.6	59.8	63.2	63.8	62.5	64.1	65.9	67.6	67.2	67.1
Oil Temp	°C	86.7	82.6	84.7	84.6	84.3	93.1	94.3	95.5	95.5	95.5
Engine Water Temp Out	°C	73.5	68.6	70.8	70.3	69.6	75.7	74.4	73.1	73.2	74.5
Fuel Flow Rate In (Corrected)	L/min	2.3	2.3	2.3	2.3	2.3	2.4	2.4	2.4	2.4	2.4
Exhaust Temp	°C	294.0	209.0	191.0	182.0	174.0	383.0	344.5	306.0	290.0	280.0
Inlet Air Flow Rate	L/min	3410	2979	2641	2520	2448	4252	3901	3549	3300	2900
Ambient Temperature	°C	25.3	25.0	24.4	24.4	24.5	32.5	33.6	34.6	35.2	35.5
Ambient Air Pressure	mmHg	766.2	766.2	766.2	766.2	766.2	734.2	734.2	734.2	734.2	734.2
Fuel Temp	°C	36.7	35.9	33.8	34.3	34.6	39.9	40.8	41.7	41.9	42.1
Fuel Flow Rate Out (Corrected)	L/min	2.1	2.2	2.2	2.2	2.3	2.1	2.2	2.2	2.3	2.3
Exhaust Analysis Measurements											
CO	%	0.24	0.17	0.10	0.10	0.11	0.26	0.18	0.11	0.11	0.11
CO2	%	8.43	6.13	3.82	3.03	2.80	8.90	6.78	4.66	3.32	3.01
NOX	ppm	700	458	333	255	223	757	442	308	232	193
NO	ppm	615	320	220	143	121	700	350	197	134	105
THC (Total Hydro Carbon)	ppm	83	85	88	93	96	97	115	121	124	128
O2	%	6.8	10.3	13.8	14.3	14.5	6.1	9.3	12.5	13.5	13.8
Physical Values	**Units**										
Number of cylinders		4	4	4	4	4	4	4	4	4	4
Engine Capacity	cm3	1997	1997	1997	1997	1997	1997	1997	1997	1997	1997
Number of power strokes per revolution		2	2	2	2	2	2	2	2	2	2
Fuel Density	Kg/m3	864	864	864	864	864	864	864	864	864	864
Cooling Water Density	Kg/m3	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000
Specific enthalpy of Combustion LHV of diesel fue	KJ/Kg	4.50E+04	4.50E+04	4.50E+04	4.50E+04	4.50E+04	4.50E+04	4.50E+04	4.50E+04	4.50E+04	4.50E+04
Specific heat capacity of water Cp	KJ/kgK	4.18	4.18	4.18	4.18	4.18	4.18	4.18	4.18	4.18	4.18
Specific Heat capacity of exhaust Cp	KJ/kgK	1.15	1.15	1.15	1.15	1.15	1.15	1.15	1.15	1.15	1.15
Calculated Values	**Units**										
Vs	m3	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
BMEP	kPa	289.347	190.721	79.6847	47.0271	28.0856	285.428	185.099	84.77	43.458	21.9973
Air mass flow rate	kg/s	0.06759	0.05911	0.05251	0.0501	0.04865	0.07887	0.07209	0.06537	0.06067	0.05326
Fuel mass flow rate	kg/s	0.03365	0.03358	0.03336	0.03352	0.03354	0.03406	0.03411	0.03416	0.03413	0.03413
A/F		2.01E+00	1.76E+00	1.57E+00	1.49E+00	1.45E+00	2.32E+00	2.11E+00	1.91E+00	1.78E+00	1.56E+00
BSFC		0.00076	0.00115	0.00273	0.00466	0.0078	0.00064	0.00099	0.00216	0.00421	0.00832
Cooling water mass flow rate	kg/s	0.53333	0.53333	0.53333	0.53333	0.53333	0.65	0.65	0.65	0.65	0.65
Energy Calculation	**Units**										
Rate of Energy Input (from fuel combustion)	kW	1.51E+03	1.51E+03	1.50E+03	1.51E+03	1.51E+03	1.53E+03	1.53E+03	1.54E+03	1.54E+03	1.54E+03
Rate of Energy into engine cooling water	kW	28.7584	19.6181	16.9429	14.4907	15.8283	31.5172	23.2304	14.9435	16.302	20.1058
Rate of Energy into Exhaust	kW	31.285	19.6125	16.4526	15.155	14.1307	45.5161	37.9761	31.0621	27.7768	24.5717
Calculated Losses %	**Units**										
Brake Power	%	2.93E+00	1.93E+00	8.13E-01	4.77E-01	2.85E-01	3.47E+00	2.25E+00	1.03E+00	5.27E-01	2.67E-01
Cooling water	%	1.90E+00	1.30E+00	1.13E+00	9.61E-01	1.05E+00	2.06E+00	1.51E+00	9.72E-01	1.06E+00	1.31E+00
Exhaust	%	2.07E+00	1.30E+00	1.10E+00	1.00E+00	9.36E-01	2.97E+00	2.47E+00	2.02E+00	1.81E+00	1.60E+00
Other losses	%	9.31E+01	9.55E+01	9.70E+01	9.76E+01	9.77E+01	9.15E+01	9.38E+01	9.60E+01	9.66E+01	9.68E+01

Figure 3 Excel spreadsheet showing measured values of experiment in yellow and blue. And calculated values including energy calculation values and calculated losses in green.

Experiment setup images

Figure 4 Experimental engine setup.

Figure 5 Computer displaying various experimental values such as exhaust temperature, fuel temperature and oil temp etc.

Figure 6 Fuel System Analyser which is used to accurately measure the fuel flow, fuel temperature and fuel density going to and from the engine during testing.

Figure 7 Exhaust Gas Analysis System. Analyses levels of exhaust gases such as CO, CO2, O2 etc.

Calculations:

Speed 1 TEST A

$V_s\ (m^3) = \text{Engine Capacity}\ (cm^3) / (n.1 \times 10^6 (cm^3/m^3))$

$V_s\ (m^3) = \dfrac{1997 / 1 \times 10^6}{2}$

$V_s = 9.985 \times 10^{-4}\ m^3$

$= 9.99 \times 10^{-4} (m^3)$ (3sf)

$Bmep\ (kPa) = \dfrac{P_b\ n_R}{N_n\ V_s}$

$Bmep\ (kPa) = \dfrac{44.3 \times 1}{(2300 \div 60) \times 4 \times 9.99 \times 10^{-4}}$

$Bmep = 289.2022457$

$Bmep = 289 (kPa)$ (3sf)

$\dot{m}_a = \dot{V}_a \rho_a$ (kg/s) Air flow mass rate

$\rho_a = \dfrac{P_a}{R T_a}$

$\rho_a = \dfrac{766.2 \times 133}{287.1 \times (25.3 + 273.15)}$

$\rho_a = 1.189293411 \approx 1.19$ (3sf)

$\dot{m}_a = \left(\dfrac{3410 \times 10^{-3}}{60}\right) \times 1.19$

$\dot{m}_a = 0.06763166667$

$\dot{m}_a = 0.0676$ (Kg/s) (3sf)

Figure 8 Handwritten calculations of swept volume, brake mean effective pressure and air mass flow rate. Results highlighted in red and reported to three significant figures.

$\dot{m}_f\ (kg/s) = \dot{V}_f \times \rho_f$ fuel mass flow rate

$$\dot{m}_f = \left(\frac{2.337}{1000 \times 60}\right) \times 864$$

$$\dot{m}_f = 0.0336528$$

$$\dot{m}_f = 0.0337\ (kg/s)\ (3sf)$$

$$A/F = \frac{\dot{m}_a}{\dot{m}_f}$$

$$A/F = \frac{0.0676}{0.0337}$$

$$A/F = 2.005934718$$

$$A/F = 2.01\ (3sf)$$

$$BSFC\ (kg/kJ) = \frac{\dot{m}_f}{P_b}$$

$$BSFC = \frac{0.0337}{44.3}$$

$$BSFC = 7.607223476 \times 10^{-4}$$

$$BSFC = 7.61 \times 10^{-4}\ (kg/kJ)\ (3sf)$$

$\dot{m}_w\ (kg/s) = \dot{V}_w \times \rho_w$ water mass flow rate

$$\dot{m}_w = \left(\frac{32}{1000 \times 60}\right) \times 1000$$

$$\dot{m}_w = 0.5\dot{3}$$

$$\dot{m}_w = 0.533\ (kg/s)\ (3sf)$$

Figure 9 Handwritten calculations of fuel mass flow rate, air/fuel ratio, brake specific fuel consumption and cooling water mass flow rate. Results highlighted in red and reported to three significant figures.

$\dot{Q}\,(kW) = \dot{m}_f \times h_{LHV}$ Rate of energy input (fuel comb)

$\dot{Q} = 0.0337 \times 4.5\times10^4$

$\dot{Q} = 1516.5$

$\dot{Q} = 1520\,(kW)$ (3sf)

$\dot{Q}_w\,(kW) = \dot{m}_w C_{pw} \Delta T_w$ Rate of energy into cooling water

$\dot{Q}_w = 0.533 \times 4.18 \times ((73.5 + 273.15) - (60.6 + 273.15))$

$\dot{Q}_w = 28.740426$

$\dot{Q}_w = 28.7\,(kW)$ (3sf)

$\dot{Q}_{ex}\,(kW) = \dot{m}_{ex} C_{ex} (T_{ex} - T_a)$ Rate of energy into exhaust

$\dot{Q}_{ex} = (0.0676 + 0.0337) \times 1.15 \times (294 - 25.3)$

$\dot{Q}_{ex} = 31.3022065$

$\dot{Q}_{ex} = 31.3\,(kW)$ (3sf)

$$P_b\,loss\,(\%) = \frac{P_b}{\dot{Q}}$$

$$P_{bL} = \frac{44.3}{1520} \times 100$$

$P_{bL} = 2.91447...$

$P_{bL} = 2.91\,\%$ (3sf)

Figure 10 Handwritten calculations for the engine energy balance, showing fuel energy input, cooling water loss, exhaust energy and the distribution of fuel energy into brake power. Results highlighted in red and reported to three significant figures.

$$\dot{Q}_W \text{ Loss } (\%) = \frac{\dot{Q}_W}{\dot{Q}} \times 100$$

$$\dot{Q}_{WL} = \frac{28.7}{1520} \times 100$$

$$\dot{Q}_{WL} = 1.88815\ldots$$

$$\dot{Q}_{WL} = 1.89\ \% \quad (3sf)$$

$$\dot{Q}_{ex} \text{ Loss } (\%) = \frac{\dot{Q}_{ex}}{\dot{Q}} \times 100$$

$$\dot{Q}_{exL} = \frac{31.3}{1520} \times 100$$

$$\dot{Q}_{exL} = 2.0592105\ldots$$

$$\dot{Q}_{exL} = 2.06\ \% \quad (3sf)$$

$$\text{Other Loss } (\%) = 100 - P_{bL} - \dot{Q}_{exL} - \dot{Q}_{WL}$$

$$L_o = 100 - 2.06 - 1.89 - 2.91$$

$$L_o = 93.14$$

$$L_o = 93.1\ \%$$

Figure 11 Handwritten calculations showing the distribution of fuel energy into cooling water, exhaust gases and other losses. Results highlighted in red and reported to three significant figures.

Energy Balance

2300 rpm

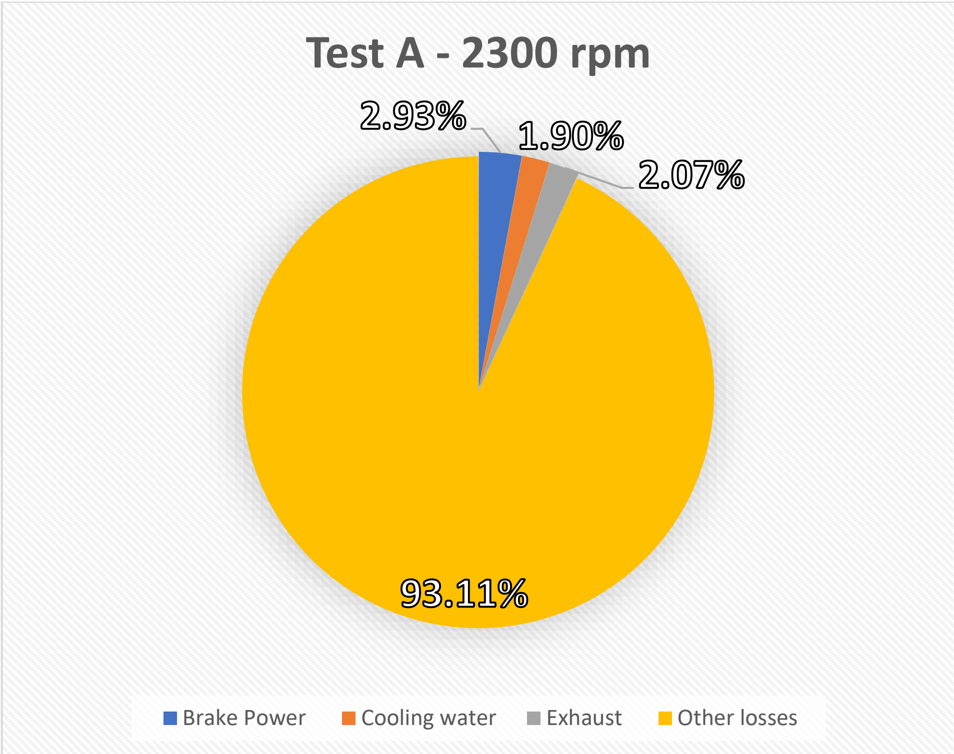

Figure 12 Energy balance showing distribution of fuel energy into brake power and losses for Test A at 2300 rpm

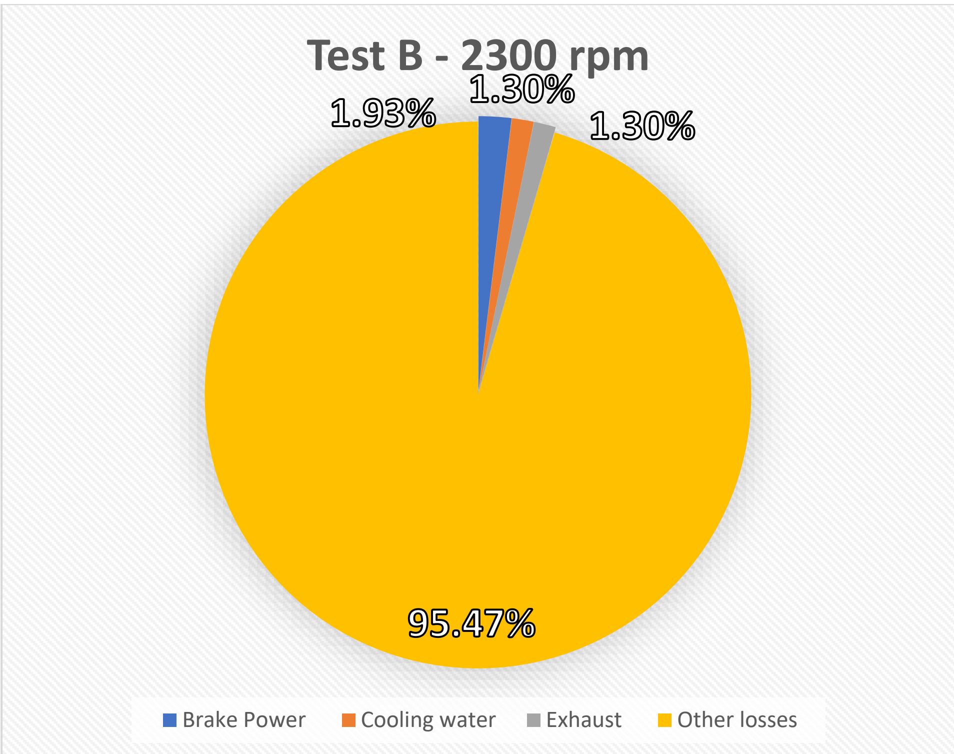

Figure 13 Energy balance showing distribution of fuel energy into brake power and losses for Test B at 2300 rpm

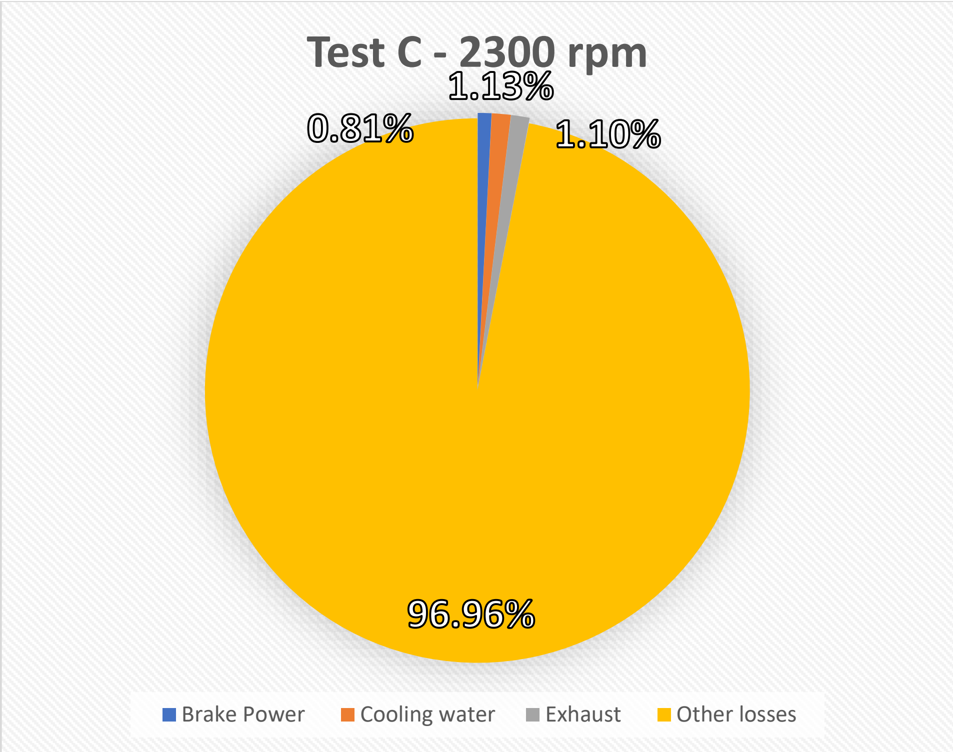

Figure 14 Energy balance showing distribution of fuel energy into brake power and losses for Test C at 2300 rpm

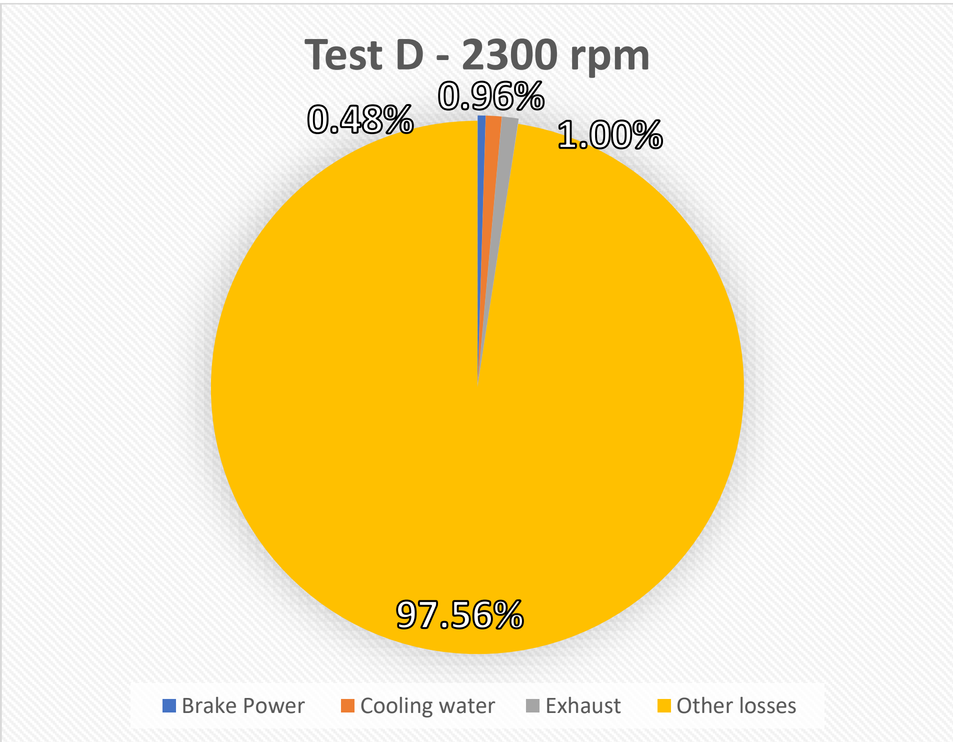

Figure 15 Energy balance showing distribution of fuel energy into brake power and losses for Test D at 2300 rpm

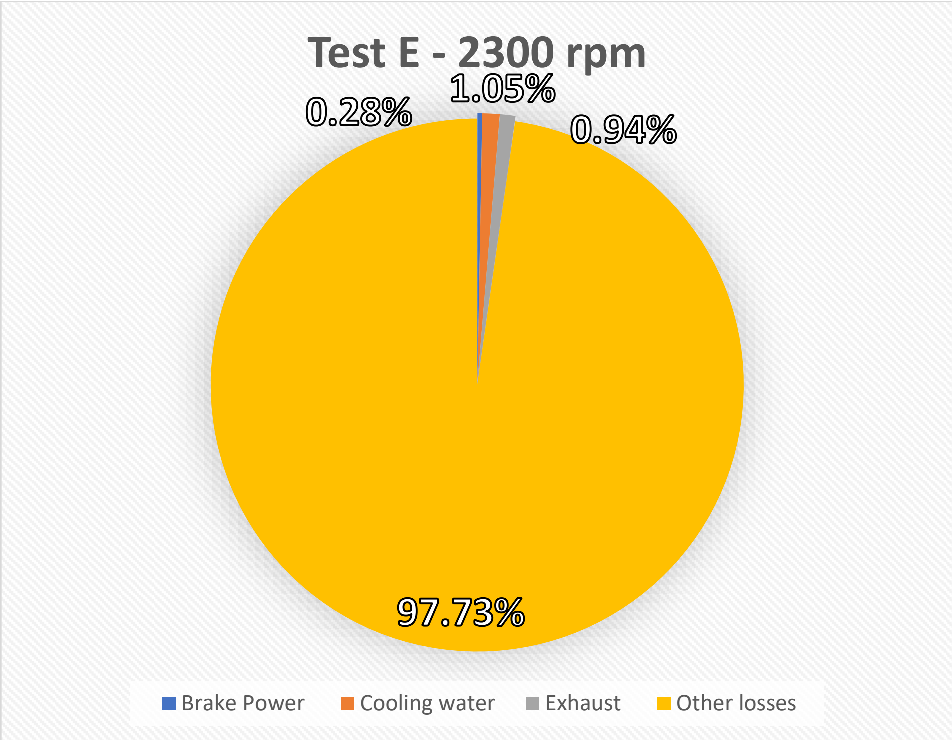

Figure 16 Energy balance showing distribution of fuel energy into brake power and losses for Test E at 2300 rpm

2800 rpm

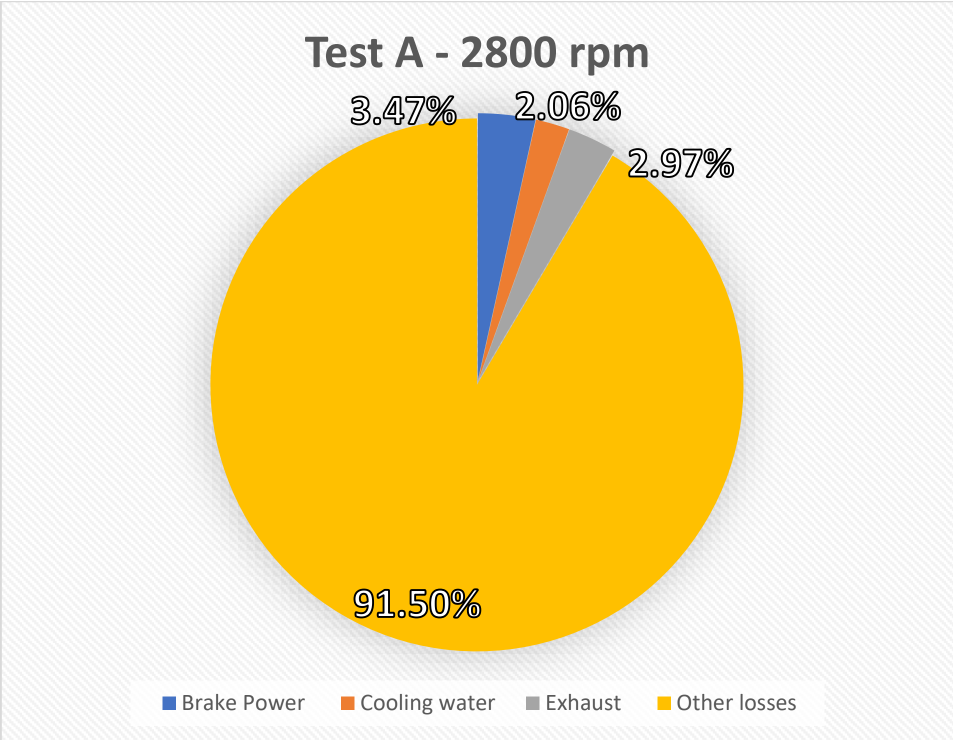

Figure 17 Energy balance showing distribution of fuel energy into brake power and losses for Test A at 2800 rpm

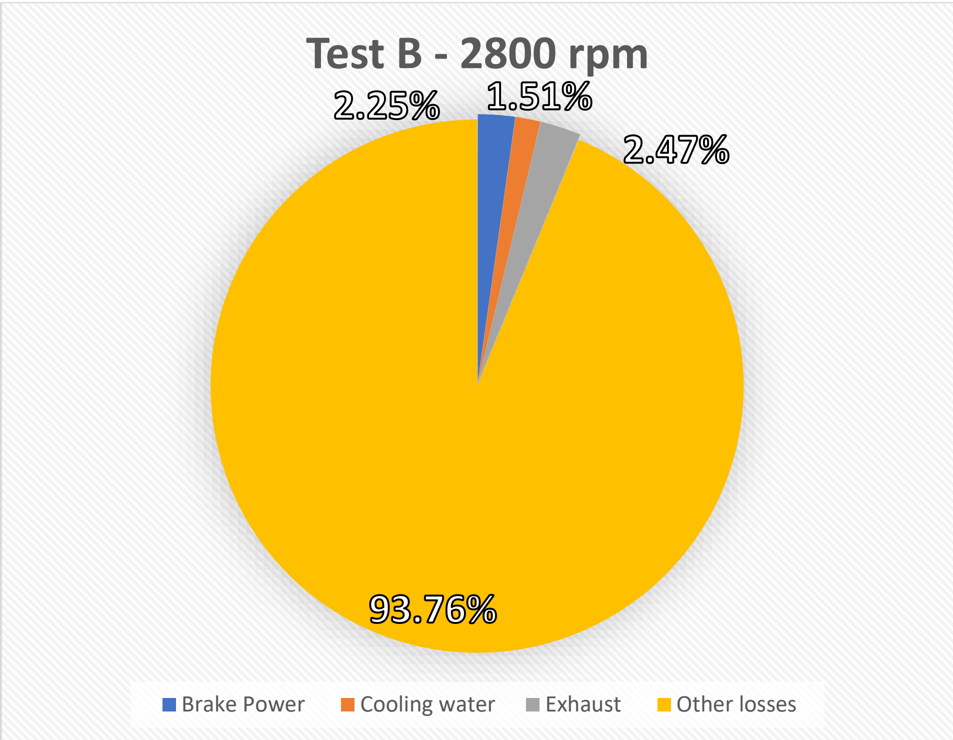

Figure 18 Energy balance showing distribution of fuel energy into brake power and losses for Test B at 2800 rpm

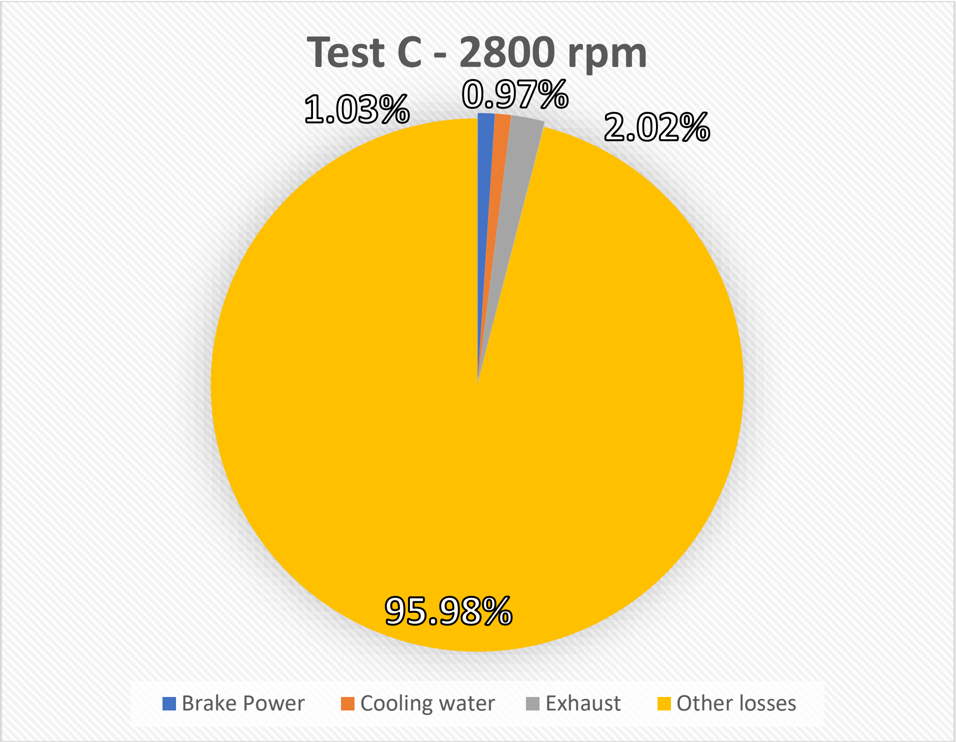

Figure 19 Energy balance showing distribution of fuel energy into brake power and losses for Test C at 2800 rpm

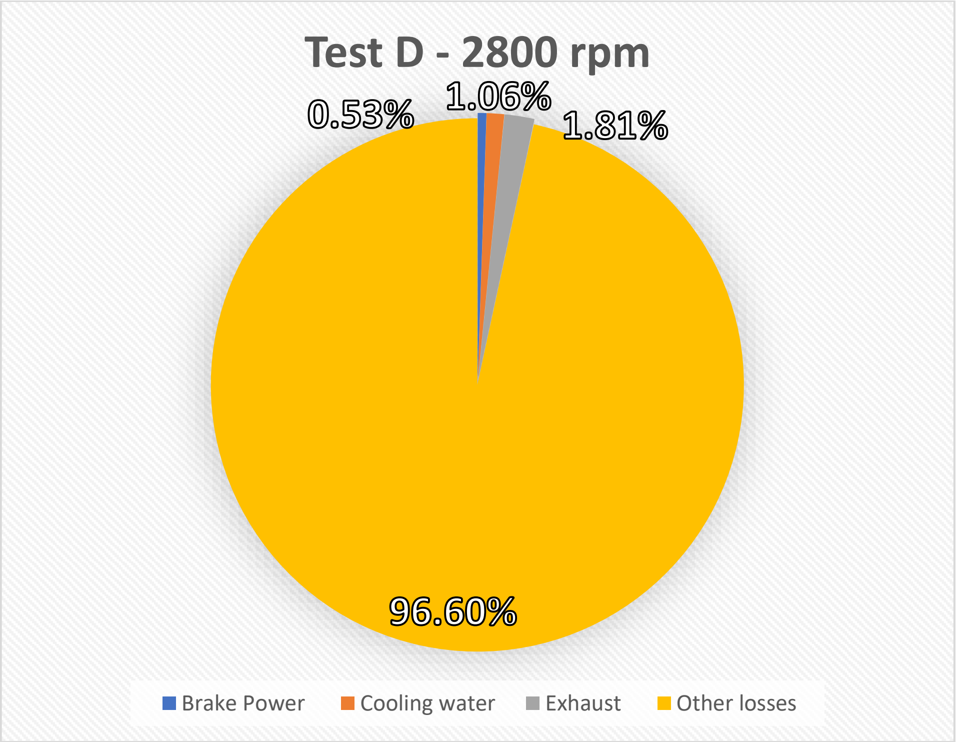

Figure 20 Energy balance showing distribution of fuel energy into brake power and losses for Test D at 2800 rpm

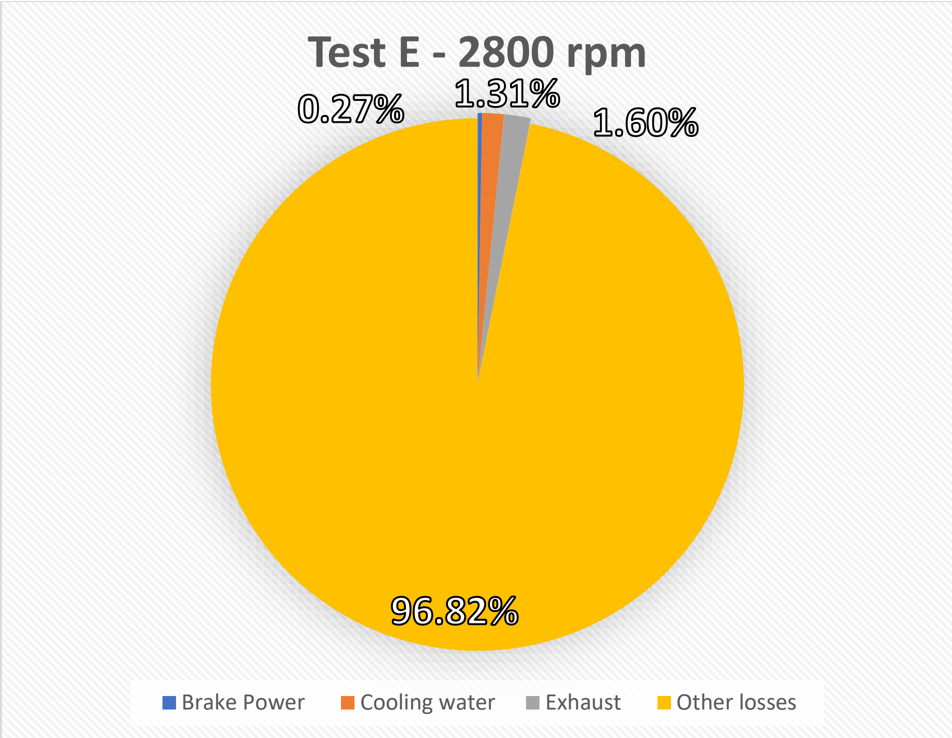

Figure 21 Energy balance showing distribution of fuel energy into brake power and losses for Test E at 2800 rpm

Discussion

The pie charts illustrate the distribution of fuel energy at selected operating points. At lower BMEP, only a small proportion of the fuel energy is converted into brake power, while a larger fraction is lost through cooling water, exhaust gases, and other losses. This behaviour is expected at low engine load, where fixed friction and pumping losses represent a significant proportion of the total energy input.

Despite operating at a lower brake mean effective pressure, the brake thermal efficiency at 2800 rpm in Test A (3.4%) (*Fig. 11*) was higher than at 2300 rpm Test A (2.9%) (*Fig. 6*). This indicates that, under the tested conditions, the engine converted a greater proportion of fuel energy into useful brake power at the higher speed. However, at extremely low BMEP, the difference in brake thermal efficiency between 2300 rpm and 2800 rpm becomes negligible. In Test E, efficiencies of (0.28%) (*Fig. 10*) and (0.27%) (*Fig. 15*) were recorded respectively, reflecting the loss dominated operation at very low engine load. This is likely linked to the increase of the A/F ratio at higher rpms (*Fig. 16*), where the fuel is used more efficiently at higher rpm due to a higher intake of air allowing for more powerful combustion using less fuel.

At both engine speeds the trend from Test A to Test E is that of declining brake thermal efficiency as BMEP decreases.

$$Brake\ Thermal\ Efficiency = \frac{1.93}{2.93} = 0.659$$

$$BMEP = \frac{190}{289} = 0.657$$

By calculating the change in brake thermal efficiency along with the change in BMEP at 2300 rpm from Test A to Test B, the values are demonstrated to be approximately identical (negligible difference). Demonstrating the directly proportional relationship between brake power and BMEP.

Brake power is proportional to torque and RPM. The torque and RPM of the 2800rpm tests seem to have stayed constant (178 Nm), this should mean a contradiction. However, the sensors used may have limited resolution. Meaning the true torque may have drifted but the sensor rounded it to the closest value it can display. And small torque drops can cause a significant change in brake power, therefore, the change in torque is not shown in the spreadsheets torque value. Instead reflected in the brake power.

Measured against BMEP

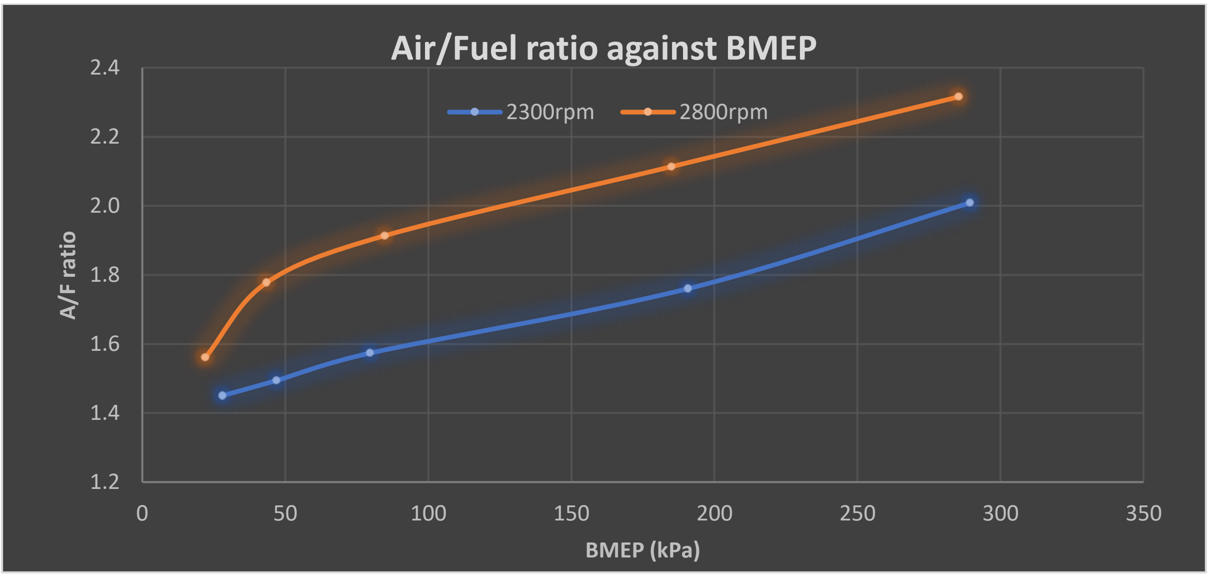

Figure 22 Variation of engine air/fuel ratio as a function of brake mean effective pressure (BMEP) at engine speeds of 2300 rpm and 2800 rpm.

At low load combustion is unstable, residual gases are high, and the ECU often runs less optimally. Improved fuel atomisation at higher engine speeds results in finer fuel droplets and enhanced air-fuel mixing, promoting more complete combustion. As BMEP increases from very low values, combustion stabilises, volumetric efficiency improves and air utilisation improves faster than fuel increase. Volumetric efficiency describes the effectiveness of cylinder filling and increases with engine speed and load due to improved intake airflow momentum and reduced relative throttling losses.

At 2800 rpm, compared to 2300 rpm: intake air velocity is much higher, in-cylinder turbulence rises sharply, and volumetric efficiency improves more rapidly with load. So, from low BMEP upwards, airflow capability improves very quickly, and fuel delivery increase more gradually. This results in the steeper initial A/F ratio increase displayed in the graph of 2800 rpm at approximately (25-50 kPa). Beyond (75 kPa) the line representing the A/F at 2800 rpm follows the 2300 rpm gradient, this explains something important about the engine; once BMEP reaches moderate values, intake flow approaches its effective limit, volumetric efficiency gains level off and fuel flow begins to scale more directly with load. At that point the same combustion constraints dominate at both speeds, the ECU A/F ratio control strategies cross over.

This means that the engine benefits from higher speed only at low-moderate load, at higher BMEP it becomes air-handling limited and no further A/F ratio advantage is gained from speed alone beyond the initial improvement region. Where there is a relatively consistent (0.4) A/F ratio difference between the two engine speeds.

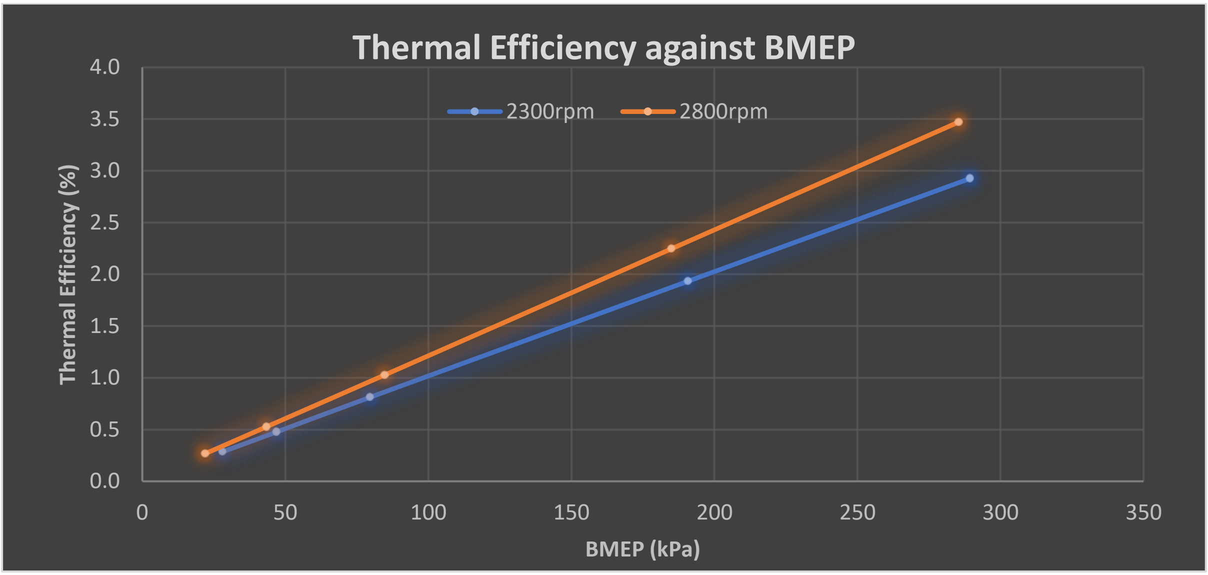

Figure 23 Variation of engine thermal efficiency (%) as a function of brake mean effective pressure (BMEP) at engine speeds of 2300 rpm and 2800 rpm.

Thermal efficiency is the useful energy converted from combustion in the engine that moves the crankshaft and runs the engine. In a real-world scenario, it would move the wheels against the load provided, simply it would move the car. The percentage is very low as previously mentioned due to the extremely low loads. As demonstrated by the graph, the efficiency increases as load increases and is better at 2800 rpm compared to 2300 rpm at the same load.

This is because at higher rpm, the combustion runs slightly leaner. Leaner mixtures reduce heat losses and fuel wasted per cycle. This is programmed by manufacturers in the ECU to avoid enrichment unless stability demands it. Furthermore, increasing engine load at constant speed increases the proportion of fuel energy converted into useful work, as fixed and pumping losses become less significant relative to brake output. Therefore, improving efficiency in this scenario.

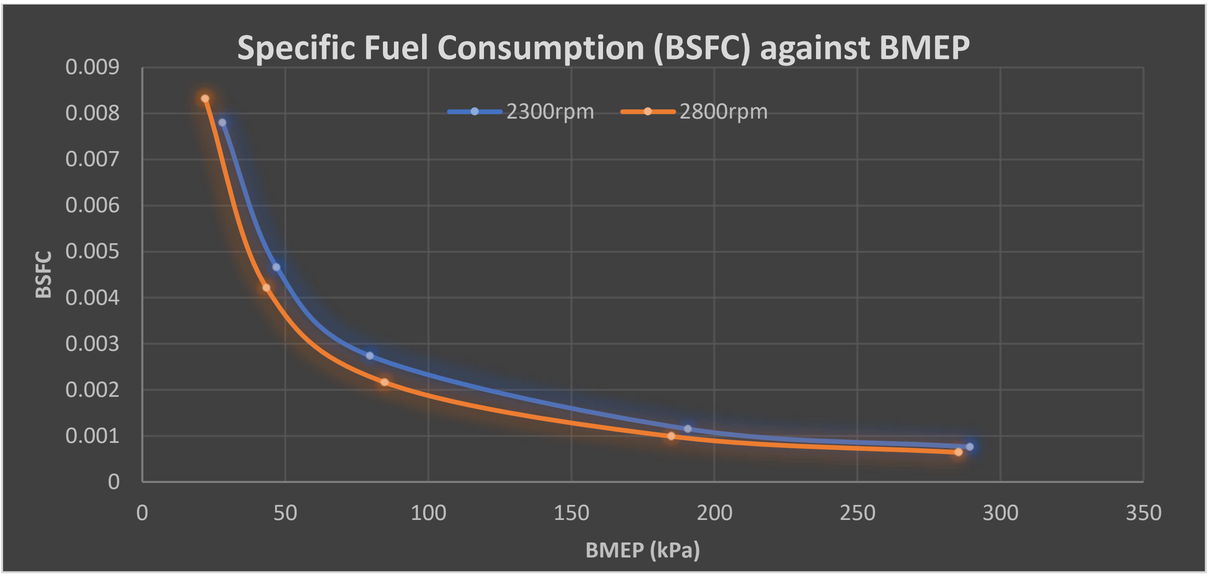

Figure 24 Variation of engine brake specific fuel consumption as a function of brake mean effective pressure (BMEP) at engine speeds of 2300 rpm and 2800 rpm.

Specific fuel consumption is fuel per unit useful power or, how much fuel is needed to provide a certain amount of power. Lower BSFC means more efficient use of fuel energy. This graph demonstrates once again that as load increases, efficiency increases accordingly, and that higher rpm is more efficient at the same load. Fuel efficiency is a direct indicator of engine efficiency.

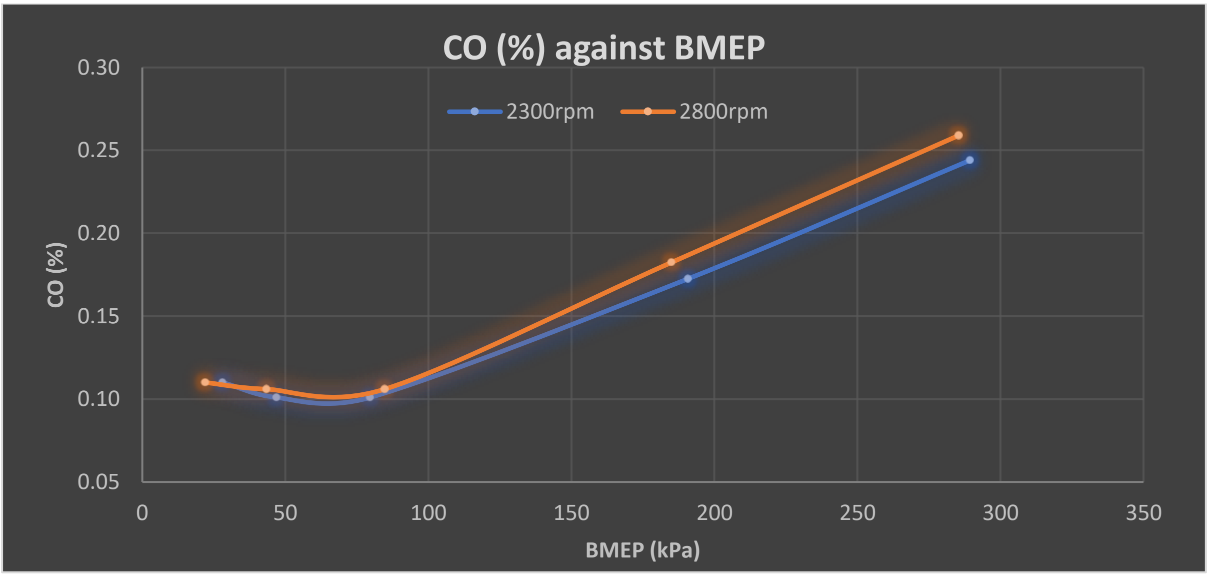

Figure 25 Variation of exhaust carbon oxide concentration (% vol) as a function of brake mean effective pressure (BMEP) at engine speeds of 2300 rpm and 2800 rpm.

This graph displays how much carbon oxide gases as a percentage concentration of the total exhaust fumes are released according to BMEP. From approximately 25 Kpa to 75 Kpa the percentage seems to slightly decrease, until past 75 Kpa where it rises linearly. Although the measured A/F ratio increases with BMEP, the rapid increase in fuel energy release and local mixture inhomogeneities limit the availability of free oxygen for complete CO oxidation. As a result, CO concentration increases at high load despite a higher global A/F ratio.

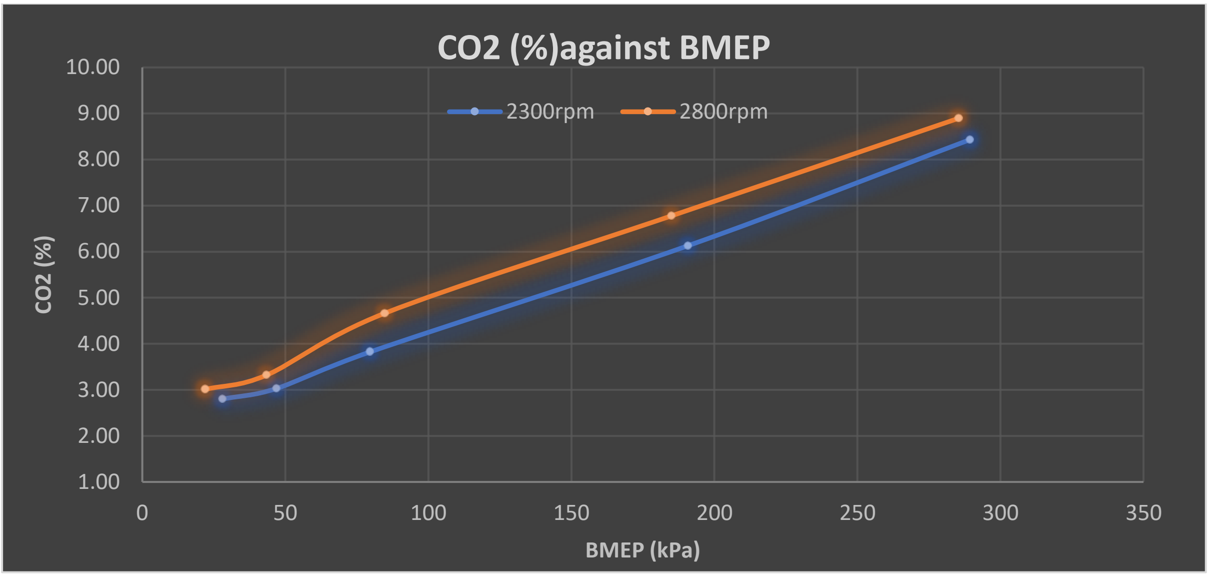

Figure 26 Variation of exhaust carbon dioxide concentration (% vol) as a function of brake mean effective pressure (BMEP) at engine speeds of 2300 rpm and 2800 rpm.

This graph also shows how much carbon dioxide gases as a percentage concentration of the total exhaust fumes are released according to BMEP. From approximately 25 Kpa to 50 Kpa, the 2800 rpm plot is close in value to 2300 rpm, but after 50 Kpa the difference increases until 75 Kpa and slightly decreases in difference throughout. This is likely because at low load the advantage of higher engine speed is lower, hence the small difference at low BMEP. CO responds strongly to enrichment and oxygen limitation; CO2 responds to overall combustion completeness. At higher BMEP, the better air fuel mixing, faster flame development and higher turbulence intensity of 2800 rpm means the gap between CO2 percentage of both engine speeds widens. As previously mentioned, CO is an intermediate product and exists only if oxidation is incomplete, whereas CO2 is the end-product, and mainly depends on how much fuel is burned and how complete combustion is overall. Which is why, unlike CO, the CO2 does not have an initial threshold for its’ value to start increasing with BMEP.

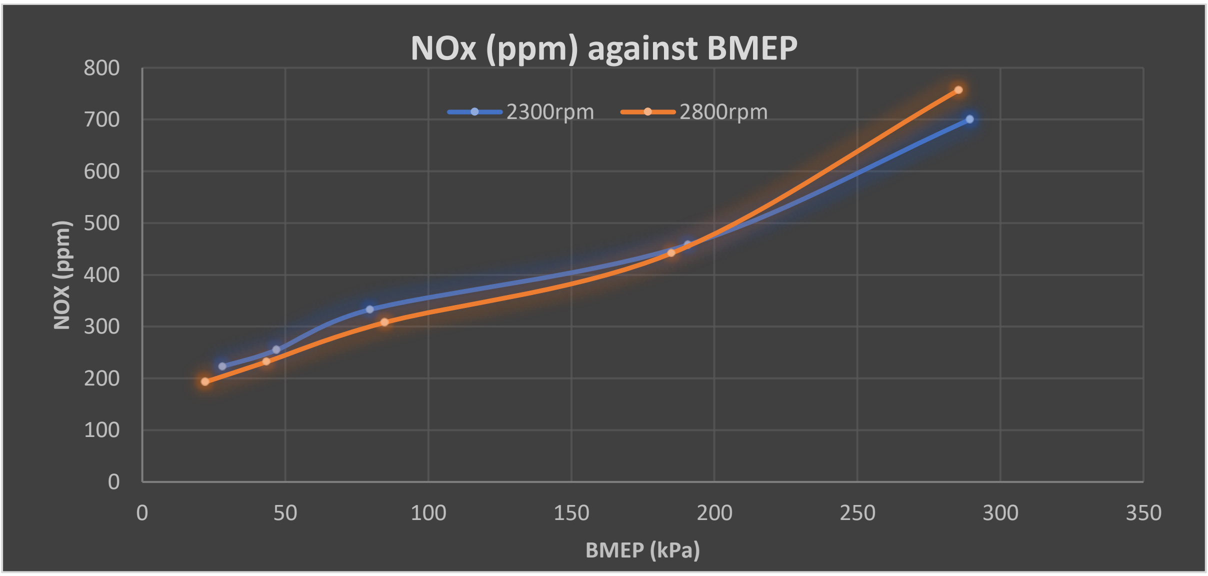

Figure 27 Variation of exhaust nitrogen oxides (ppm vol) as a function of brake mean effective pressure (BMEP) at engine speeds of 2300 rpm and 2800 rpm.

NOx formation increased non-linearly with BMEP due to its strong dependence on in-cylinder temperature, oxygen availability, and residence time. At low load, NOx remained low, particularly at 2800 rpm, due to reduced residence time at peak temperature. As BMEP increased, higher combustion temperatures promoted rapid thermal NO formation, with enhanced turbulence at 2800 rpm leading to higher peak temperatures and increased NOX compared 2300 rpm.

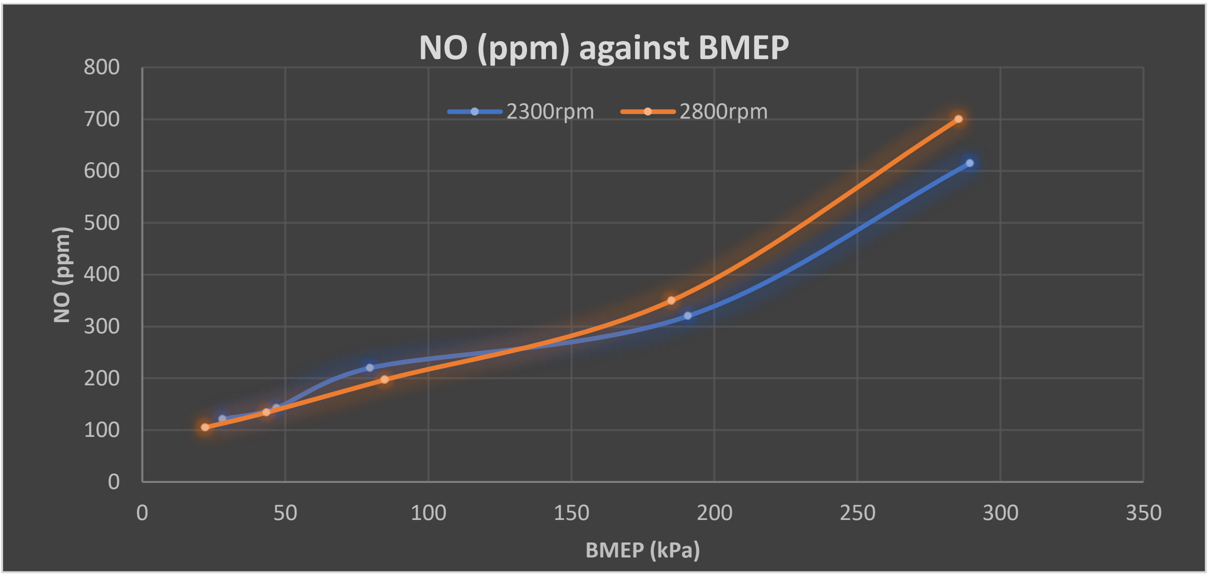

Figure 28 Variation of exhaust nitrogen oxide (ppm vol) as a function of brake mean effective pressure (BMEP) at engine speeds of 2300 rpm and 2800 rpm.

Nitric oxide (NO) is the dominant nitrogen oxide formed during high-temperature combustion within the cylinder. The term Nox refers to the combined concentration of nitrogen oxides, primarily NO and NO_2, with NO constituting most engine-out NOx emissions. The NO concentration exhibits a similar non-linear trend to Nox with increasing BMEP. However, the variation is more pronounced, as NO is the primary nitrogen oxide formed directly during high-temperature combustion and is more sensitive to peak in-cylinder temperature and residence time.

At higher BMEP, enhanced turbulence and faster combustion at 2800 rpm result in higher peak flame temperatures, causing NO formation to increase more rapidly than at 2300 rpm, leading to a clearer crossover between the two speeds.

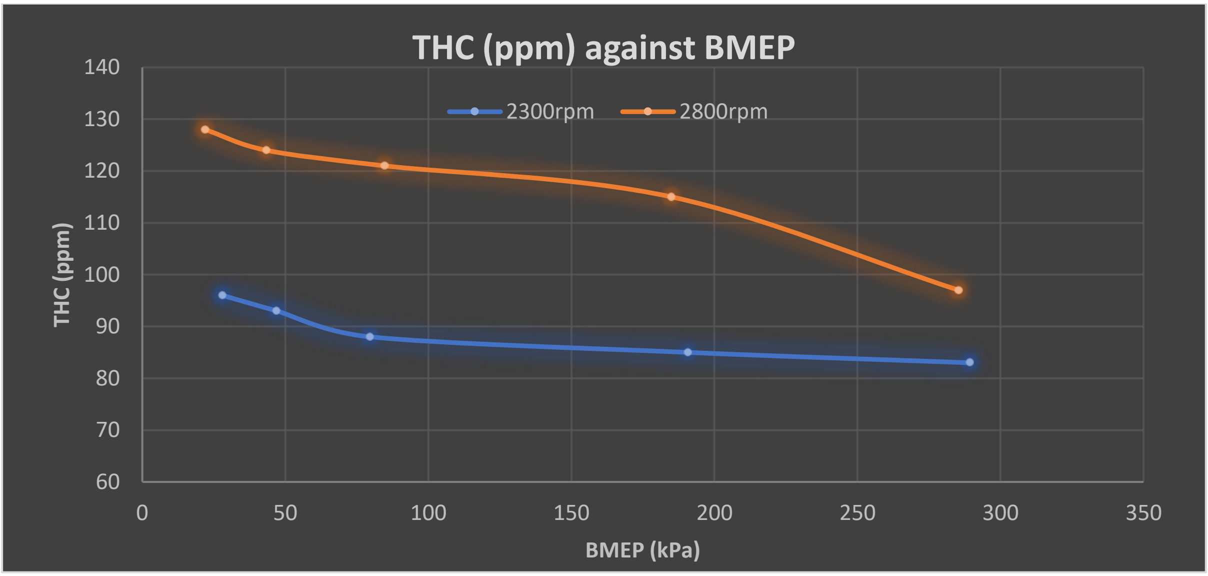

Figure 29 Variation of exhaust total hydrocarbons (ppm vol) as a function of brake mean effective pressure (BMEP) at engine speeds of 2300 rpm and 2800 rpm.

Total Hydrocarbons (THC) represent unburned or partially burned fuel molecules that exit the engine without being oxidised. THC mainly originates from crevice volumes such as piston ring grooves, wall quenching and poor mixing/misfire. The peak difference between the two engine speeds is about 30 ppm. This is significant and is linked to the higher speed leading to less time for flames to reach crevices, less time for post-flame oxidation which results in more hydrocarbons escaping and a higher THC baseline. However, towards higher BMEP, the THC numbers drop at 2800 rpm and remain relatively constant at 2300 rpm. This is because as BMEP increases there is higher turbulence, a stronger flame, higher wall temperatures and reduced quenching. So despite shorter residence time, the improvement in combustion quality dominates and THC continues to fall and reaches a clear minimum.

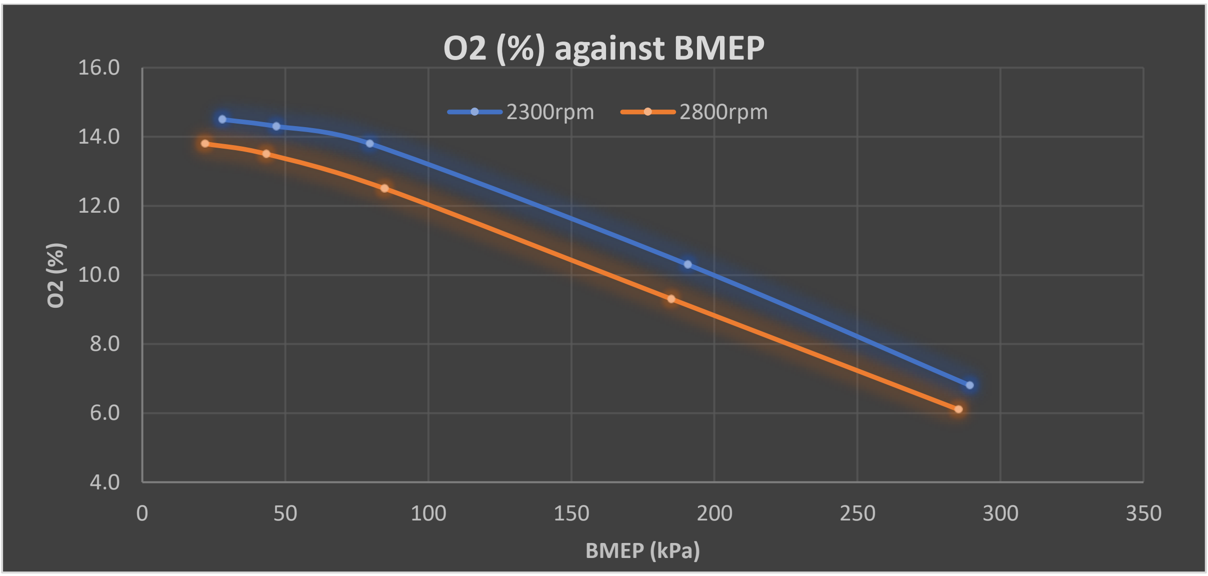

Figure 30 Variation of exhaust oxygen concentration (% vol) with brake mean effective pressure at 2300 rpm and 2800 rpm.

O_2 is the fraction of oxygen that entered the engine but was not consumed during combustion, which signifies how much excess air there was and whether local combustion was lean, stoichiometric or rich.

O_2 decreased as BMEP increased because more fuel is injected per cycle, more oxygen is required for combustion and therefore, a larger fraction of the intake oxygen is consumed. O_2 is lower at 2800 rpm than at 2300 rpm, this is likely due to higher turbulence intensity, better air-fuel mixing and faster fuel propagation; this results in more effective use of oxygen and less oxygen surviving to the exhaust. Furthermore, 2800 rpm reaches higher peak flame temperatures and oxidation reactions proceed further leading to higher O_2 consumption and less residual oxygen.

Conclusions:

This experiment investigated the performance and emissions of a compression ignition engine operating at 2300 rpm and 2800 rpm under different load conditions. Engine load was represented using BMEP, allowing direct comparison between test points at both speeds.

The results showed that increasing BMEP improved brake thermal efficiency and reduced BSFC at both engine speeds. This indicates that the engine operated more efficiently a higher load, as a greater proportion of the fuel energy was converted into useful work. At very low BMEP, thermal efficiency dropped significantly, showing that the engine operated inefficiently under light load conditions.

Exhaust emissions were also affected by load and speed. Higher emissions of CO and THC were observed at low BMEP, suggesting incomplete combustion due to low in-cylinder temperatures. As BMEP, increased, CO_2 levels rose and oxygen levels decreased, indicating more complete combustion. NO_x showed a non-linear trend with load, reflecting changes in combustion temperature and residence time.

Overall, the experiment demonstrated that engine load has a strong influence on diesel engine efficiency and emissions, and that operating the engine at higher loads results in improved performance and cleaner combustion.